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#	Source Table/Variable	Situation Requiring Resolution	Action	Reasoning / Comments	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)					
1	Days to months conversion	Assuming 30.4375 day per month	3 months = 91 days, 6 months=183 days, 9 months=274 days, 12 months=365 days The team will additionally include patients who had ≤1 non-diagnostic medical claim for MM during the baseline period. These patients were excluded by Optum originally. If a patient has newly diagnosed or active MM, it would be likely that they would experience frequent activity, hence why we look for 2 non-diagnostic MM claims close together in the first step of the attrition table. A smoldering patient would have infrequent activity, hence why a claim for MM that falls outside of the 30, 60 or 90 days window requirement that's used to identify newly diagnosed patients is likely a smoldering patient.	Consistent with derivation other GSK Oncology studies		OK19012026					
2	ELUG_COH	Study team does not want to exclude smoldering MM patients. Optum built study cohort with the intention to exclude smoldering MM patients.		Study team will consider excluding patients with a stand alone treatment code (rather than any non-diagnostic code) on an MM claim in the baseline period for future sensitivity analysis. This would not remove smoldering patients but rather ensure our patients were newly active MM patients		OK19012026					
3	FU_PD	Original IE criteria from Optum only considers patients with at least 6 mos of CE from index onwards.	Study team wants to keep criteria as broad as possible.	Will only require at least a day of CE post index, though will flag patients with 3 or more months of CE, to potentially subset on these patients for later consideration.		OK19012026					
4	MED_COND	Conditioning medications are included in MM-approved code list	Ignore these treatments for now	The team is not particularly interested in conditioning therapy. These treatments potentially will be added back to the LOT at a later time.		OK19012026					
5	Identifying inpatient and non-inpatient claims	Approach 1: Use TOS_CD if conditions are not restricted on inpatient services associated with a hospitalization which requires an admission and discharge date. 1. Extract the records from T_Medical and T_CONFINEMENT datasets Inpatient → POS IN (21, 51, 61) or TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF INPWS, FAC_IP SNF) Outpatient → none of the above Approach 2: For anything related to services happening during a hospitalization with room and board, apply the condition on the CONF_ID because the majority of the work we do includes both diagnoses and medical services e.g. procedures, and when we talk about inpatient utilization, it is usually in the context of an admission and discharge. 1. Inpatient records should be restricted to cases where CONF_ID is not NULL from the T_CONFINEMENT table, where this has associated admission and discharge dates. 2. The confinement table has summary information, so if users are pulling medical claims for specific inclusion, user should match on the CONF_ID variable, where it is not NULL in the confinement table. All other records without a CONF_ID or where CONF_ID is NULL should be considered non-inpatient. Defining mtx periods is quite complex, especially with the elimination of the 90 day gap rule originally proposed by Optum. Without allowable (90d) gaps, for which mtx periods relied on for defining starts and stops, and due to the messy nature of claims and prescriber behavior, it is very difficult to create an	Approach: We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum COM business rules guidance. Approach 1 (POSTOS). Inpatient if POS IN (21, 51, 61) OR TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF INPWS, FAC_IP SNF) Approach 2 (CONF_ID). Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both ADMIT_DATE and DISCH_DATE present Non-inpatient: All records not meeting either approach above. A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters.	Rationale: Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data: Total patients (Step 0): 94951 Any inpatient event: 43439 (45.2%) Via POSTOS (Appr 1): 42618 (44.3%) Via CONF_ID (Appr 2): 40357 (42.0%) Both approaches: 39536 (41.1%) POSTOS only: 3082 (3.2%) CONF_ID only: 821 (0.9%) Outpatient only: 52757 (54.8%)	OK19012026						
Defining maintenance periods		For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapies/therapy combinations.	Reporting mtx is not of importance/relevance to upcoming studies planned in Multiple Myeloma. Flags will enable users to define mtx themselves in the future if so needed.								